FINANCE I

Facultad de Economía y Negocios
Universidad de Chile

Prof.: Alejandro Bernales

Solemne

Time Allowed: 1.5 Hours

Instructions:

The solemne has 3 questions, each of them has the same weight.

PLEASE WRITE YOUR ID NUMBER (RUT OR PASSPORT) IN ALL THE PAGES.

PLEASE ONLY USE THE SPACE IN EACH QUESTION TO ANSWER ALL SECTIONS (USE ONLY ONE SIDE OF THE PAGE - ONE LANGUAGE)

The space provided is sufficient to give a concise and pertinent answer to each question and/or sub-point of a question.

Extra answer booklets will not be provided – please use the spaces provided only. Answers written outside the spaces or on other booklets will not be marked.

Electronic calculators may be used provided that they cannot store text

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Instrucciones:

La Solemne tiene tres preguntas, cada una de ellas tiene la misma ponderación.

POR FAVOR ESCRIBA SU RUT (O # PASAPORTE) EN TODAS LAS HOJAS.

POR FAVOR UTILICE EL ESPACIO PROVISTO EN CADA PREGUNTA PARA RESPONDER CADA SECCIÓN (USAR UN SOLO LADO DE LA HOJA - UN SOLO LENGUAJE).

El espacio entregado es suficiente para entregar una respuesta concisa y pertinente a cada pregunta y/o sub-pregunta.

No se entregarán hojas adicionales para responder por favor utilice solamente el espacio entregado. Respuestas escritas fuera de este espacio o en otras hojas no serán revisadas.

Calculadoras electrónicas pueden ser utilizadas salvo las que puedan almacenar texto

Question I. Assume a financial market in which returns are described by a two-factor model (2-factor APT). Additionally, assume that you have a well-diversified portfolio and perfect information. The relevant information for this scenario is presented in the following table:

Stocks	$E(r_i)$	β_1	β_2
LAN	13.5%	0.7	0.6
KO	14.7%	0.9	0.5
AA	9.9%	0.4	0.3

1a) Please, provide the second equation (as explained in class) that describes the equilibrium expected returns in the APT.

Hint. The second APT equation is:

$$E(r_i) = r_f + \sum_{j=1}^{k} \beta_{i,j} \lambda_j$$

Answer:

We make a system of equations with the data:

$$13.5\% = r_f + \lambda_1 * 0.7 + \lambda_2 * 0.6$$
$$14.7\% = r_f + \lambda_1 * 0.9 + w_2 * 0.5$$
$$9.9\% = r_f + \lambda_1 * 0.4 + \lambda_2 * 0.3$$

Finally, the values are:

$$r_f = 0.055$$
$$\lambda_1 = 0.08$$
$$\lambda_2 = 0.04$$

Therefore, the equation is:

$$E(r_i) = 0.055 + \beta_1 * 0.08 + \beta_2 * 0.04$$

Pregunta I. Supongamos un mercado financiero en el que los retornos son descritos por medio de un modelo de dos factores (APT de 2 factores). Además, considere que tiene una cartera bien diversificada e información perfecta. La información relevante para esta pregunta se encuentra en la siguiente tabla:

Stocks	$E(r_i)$	β_1	β_2
LAN	13.5%	0.7	0.6
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1a) Encuentre la segunda ecuación (como fue explicado en clase) que describe los retornos esperados de equilibrio en el APT.

Hint. La segunda ecuación del APT es:

$$E(r_i) = r_f + \sum_{j=1}^{k} \beta_{i,j} \lambda_j$$

Respuesta

1b) Suppose that the stock PG has the following features:
Expected return: 15%, $\beta_1 = 1.1$; $\beta_2 = 0.8$.
Does the stock PG have abnormal returns under this APT model?

Answer

The APT equation is:

$$E(r_i) = 0{,}055 + \beta_1 * 0{,}08 + \beta_2 * 0{,}04$$

We replace the values of β_1 and β_2, and we got:

$$E(r_i) = 0{,}055 + 1{,}1 * 0{,}08 + 0{,}8 * 0{,}04$$

$$E(r_i) = 17{,}5\% > 15\%$$

The PG stock return is abnormal under this APT model with an alpha value of -2.5%.

1b) Suponga que el stock de PG tiene las siguientes características:
Retorno esperado: 15%, $\beta_1 = 1.1$; $\beta_2 = 0.8$.
¿El stock de PG tiene retornos que son anormales bajo este modelo APT?

Respuesta

Question II:

2a) In the course, we were discussing three ways in which some investors can have more information than others. First, in an illegal way, some investors can obtain internal information from companies (through friends or relatives working within those companies). Second, some investors may have exceptionally powerful computers and highly skilled teams, including members with PhDs in Finance, Economics, Mathematics, Programming, amongst others. In this second scenario, these sophisticated teams can extract more knowledge from publicly available information. What is the third way in which certain agents can possess more information than others? Please provide an example.

Answer:

The third way in which some agents can possess more information than others is related to the learning process and the attention of agents. Naturally, learning new information (after market events that affect investments) takes time. In addition, investors sometimes are not fully focused on monitoring their investments because they have other tasks to attend to. Consequently, due to the time required for the learning process and the inattention of investors, some quicker investors can gain access to new information more rapidly and use it to their advantage against slower-learning and inattentive investors.

For example, an investor with investments who is going for lunch may miss new information regarding such investments, and quick investors can exploit the inattentive investor's investment positions.

2b) In class, we discussed the concept "walk up the book", in relation to some illegal practices of some managers and accountants, where they modify some numbers of the accounting book of firms. Please, provide an example of such illegal practices in companies.

Answer

This sentence is not true, since the concept of "walking up the book" is not related to the accounting book of firms (and it is not illegal). This concept is related to trading, when a large order is sent to be traded instantaneously in the market (e.g., a large market order in limit order markets), where this large order is traded at different prices. This occurs because there are not enough limit orders waiting at a single price to execute such a large order. Thus, this large order is "walking" at different prices in the limit order book.

Pregunta II:

2a) En el curso, estuvimos discutiendo tres formas en las que algunos inversionistas pueden tener más información que otros. Primero, de manera ilegal, algunos inversionistas pueden obtener información interna de las empresas (a través de amigos o familiares que trabajan en esas empresas). Segundo, algunos inversionistas pueden contar con computadoras excepcionalmente potentes y equipos altamente capacitados, incluyendo miembros con doctorados en Finanzas, Economía, Matemáticas, Programación, entre otros. En este segundo escenario, estos equipos sofisticados pueden extraer más conocimiento de la información disponible públicamente. ¿Cuál es la tercera forma en que ciertos agentes pueden poseer más información que otros? Por favor, proporciona un ejemplo.

Respuesta

2b) En clase, discutimos el concepto de "caminar sobre el libro" (i.e. "walk up the book"), en relación con algunas prácticas ilegales de ciertos gerentes y contadores, donde modifican ciertos números del libro contable de las empresas. Por favor, proporcione un ejemplo de tales prácticas ilegales en las empresas.

Respuesta

Question III. Consider two bonds. The first bond (Bond I) has a face value of $100 and pays annual coupons of 8% (of face value) over 10 years, and currently it has a price at $90. The second bond (Bond II) also has a face value of $100, but pays annual coupons of 4% (of face value) over 10 years, and currently it has a price at $80. What is the 10-year zero-coupon rate? Express the 10-year zero-coupon rate in continuous time (annual base).

Answer:

Bond I

$$90 = 8e^{-1\cdot r_1} + 8e^{-2\cdot r_2} + 8e^{-3\cdot r_3} + \cdots + 108e^{-10\cdot r_{10}}$$

Bond II

$$80 = 4e^{-1\cdot r_1} + 4e^{-2\cdot r_2} + 4e^{-3\cdot r_3} + \cdots + 104e^{-10\cdot r_{10}}$$

Then, today (at $t = 0$) I can buy 2 bonds with coupons of 4% (i.e. to buy 2 Bonds II), and simultaneously sell 1 bond with coupons of 8% (i.e. sell 1 Bond I).

Then, I have a cash flow today (at t=0) of:

$$90 - 2 \cdot 80 = -70$$

Consequently, after 10 years at t=10, I will have a a chash flow (in present value) of:

$$2 \cdot 104e^{-10\cdot r_{10}} - 108e^{-10\cdot r_{10}} = 100e^{-10\cdot r_{10}}$$

This is because the coupons between t=0 and t=10 are eliminated due to the coupons' structure

This means that I am investing $70 at the zero coupon rate, to obtain a future cash flow of $100e^{-10\cdot r_{10}}$ after 10 years.

Thus,

$$100e^{-10\cdot r_{10}} = 70$$

Then,

$$r_{10} = \frac{1}{10} \cdot \ln\left(\frac{100}{70}\right) = 0.0357$$

Finally,

$$r_{10} = 3.57\%.$$

Pregunta III. Considere dos bonos. El primer bono (Bono I) tiene valor cara $100 y paga cupones anuales del 8% anual (del valor cara) a 10 años, el cual se vende actualmente en $90. El segundo bono (Bono II) también tiene un valor cara de $100, pero paga cupones anuales del 4% anual (del valor cara) a 10 años, y se vende actualmente en $80. ¿Cuál es la tasa cero cupón a 10 años? Exprese la tasa cero cupón a 10 años, en tasa continua con base anual.

Respuesta: